

DNA Double Helix and Chromosome Compaction
Astonishing Parallels to T0-Torus Geometry From Molecular
Winding to Highest Information Density

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Abstract

This paper examines the structural parallels between the DNA double helix, its hierarchical compaction into chromosomes, and the 4D torsional structure of T0 theory. The analysis shows: Both systems can be described by **the same geometric strategy – double helices winding around tori, which in turn fold hierarchically** – to store maximum information in minimum volume. The study describes **ten parallels**: (1) **Double helix as basic structure**, (2) **Winding numbers determine properties**, (3) **Hierarchical compaction across levels**, (4) **Toroidal geometry at each level**, (5) **Singularity avoidance through minimum radii**, (6) **Information maximization with volume minimization**, (7) **10,000-fold compression without loss**, (8) **Fractal self-similarity**, (9) **Topological stability**, (10) **Dynamic unfolding when needed**. The parallels suggest that DNA compaction is a **biological solution to a geometric problem** that, in the T0 description, also structures physics across many scales; they are to be read as a structural analogy, not as proof of a common mechanism.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. Here this applies in particular to the cosmic scales; the current corpus deliberately sets the cosmic sector aside, since the Standard Model does not derive it either (P39).

.1 Introduction: The Packaging Problem

DNA: 2 Meters in 6 μm

Every human cell faces a considerable geometric problem:

How does one pack ~2 meters of DNA into a nucleus of ~6 μm diameter?

This corresponds to a linear ratio of $2\text{ m}/6\text{ }\mu\text{m} \approx 3.3 \times 10^5$; the compaction per chromosome is ~10,000.

T0: Universal Information in Space

T0 theory faces an analogous problem:

How does one encode maximum physical information in finite space without singularities?

The Common Solution

The Universal Principle

Both can be described by the same geometric strategy:

Double helices → wind around **tori** → which **fold hierarchically** → and **dynamically unfold** when needed

This strategy is an **efficient solution for information storage**.

.2 The DNA Hierarchy

Level 1: The Double Helix (Molecular)

Structure:

- Two antiparallel polynucleotide strands
- Right-handed helix
- Turn: 360° per 10.5 base pairs
- Diameter: ~2 nm
- Pitch: ~3.4 nm per turn

Geometry:

$$\text{Winding number } w = \frac{n_{\text{base pairs}}}{10.5} \approx \frac{L}{3.4 \text{ nm}} \quad (1)$$

Level 2: Nucleosomes (Histones)

Structure:

- DNA wraps 1.65 times around histone octamer
- Histone core diameter: ~11 nm
- 147 base pairs per nucleosome
- "Beads on a string"

Compression: ~6-fold

Geometry – torus:

$$R_{\text{Histone}} \approx 5.5 \text{ nm}, \quad r_{\text{DNA}} \approx 1 \text{ nm} \quad (2)$$

The DNA forms a **toroidal loop** around the histone core.

Level 3: 30-nm Fiber (Solenoid)

Structure:

- Nucleosome chain folds into **solenoid**
- 6 nucleosomes per turn
- Diameter: ~30 nm
- "Fiber of fibers"

Compression: ~40-fold (cumulative)

Geometry – helix of tori

Level 4: Higher Loops (~300 nm)**Structure:**

- 30-nm fiber forms loops
 - Loops attached to protein scaffold
 - Diameter: ~300 nm
- Compression:** ~400-fold (cumulative)

Level 5: Condensed Chromatin**Structure:**

- Further folding of loop domains
 - Diameter: ~700 nm
- Compression:** ~1,000-fold (cumulative)

Level 6: Metaphase Chromosome (Maximum Compaction)**Structure:**

- Highest condensation during cell division
 - Length: ~1–10 μm
 - Diameter: ~1 μm
 - X-shaped structure (two sister chromatids)
- Compression:** ~10,000-fold.

Per chromosome: ~4–5 cm of stretched DNA → a few μm

.3 The T0 Hierarchy**Level 1: Fundamental (Sub-Planck)****Structure:** 4D torsional crystal

- Double loop – analogous to DNA double strand
- Toroidal + poloidal circulation
- Winding number $w = n_\phi / n_\theta$
- Minimum radius: $r_{\min} \approx 19.6 \ell_P$

Level 2: Particles ($\sim 10^{-15} \text{ m}$)**Structure:** Elementary particles as torus resonances

- Electrons, quarks = stable windings
- Toroidal structure on Compton scale
- Spin from winding number

Levels 3–6: Scale-Invariant Hierarchy

Further torus structures on all scales up to cosmic **[S]**:

- Atoms $\sim 10^{-10}$ m
- Planets $\sim 10^6$ m
- Stars $\sim 10^9$ m
- Galaxies $\sim 10^{20}$ m

Compression: ~ 60 orders of magnitude with $D_f = 3 - \xi$.

.4 Ten Parallels

Parallel 1: Double Helix as Basic Structure

DNA

The **double helix** is the fundamental structure:

- Two strands wound around each other
- Right-handed
- Complementary (A-T, G-C)
- Stability through **both** strands

T0

The electron model of Williamson & van der Mark (1997) describes a **double loop** that can be read as a double helix; the T0 description takes up this structure. The parallel is a classification made in this document, not a statement by the two authors:

- Two circulations: toroidal + poloidal
- Circularly polarized field
- Winding over Compton wavelength λ_C
- Stability through **both** circulations

First Parallel

Double Circulation / Double Helix

Both use **two intertwined components**:

- DNA: Two nucleotide strands
- T0: Toroidal + poloidal flow

In both cases the **factor 2** contributes to stability.

Parallel 2: Winding Numbers Determine Properties

DNA

The **number of turns** determines:

- Helix length
- Number of base pairs
- Topological properties (linking number)

- Supercoiling behavior

Example: Plasmid with 4,000 base pairs has ~380 helix turns

T0

The **winding number** $w = n_\phi/n_\theta$ determines:

- Spin: $w = 1/2 \rightarrow$ fermions
- Spin: $w = 1 \rightarrow$ bosons
- Charge from flux quantization
- Mass from resonance

Second Parallel

Winding Number = Quantum Number

DNA	T0
Number of turns determines length	Winding number determines spin
Linking number topological	Winding number topological
Supercoiling energy	Field energy

Parallel 3: Hierarchical Compaction

DNA

6 Hierarchy Levels:

DNA strand (2 nm)	Level 1	
↓		
Nucleosomes (11 nm)	Level 2	
↓		
30-nm fiber	Level 3	
↓		
300-nm loops	Level 4	
↓		
700-nm chromatin	Level 5	
↓		
Chromosome (μm)	Level 6	

×10,000 compression

T0

~16–17 Hierarchy Levels (at a factor of ~7500 per level):

From Sub-Planck (10^{-39} m) to Cosmic (10^{26} m): 65 orders of magnitude, which at 7500 per level means about 17 levels.

Third Parallel

Both use **hierarchical folding across multiple scales**:
DNA: 6 levels, 10,000-fold compression

T0: ~16–17 levels, self-similar with $D_f = 3 - \xi$

Parallel 4: Toroidal Geometry

DNA

Torus at every level:

Level 2 (Nucleosomes): DNA wraps **1.65 times** around histone core

$$\text{Torus : } R = 5.5 \text{ nm}, \quad r = 1 \text{ nm} \quad (3)$$

Level 3 (Solenoid): Nucleosome chain forms **helix** (torus-like)

Level 4+: Loop domains attached to central axis = **toroidal arrangement**

T0

Torus on every scale (cosmic level **[S]**):

- Sub-Planck: Fundamental 4D torus
- Particles: Torus resonances
- Macro: Magnetic fields, plasmatoroids
- Cosmic: Galactic spirals, cosmic web

Fourth Parallel

The torus as a recurring geometry

Why? Because it:

- Is closed (no boundaries)
- Enables two independent circulations
- Stores energy/information efficiently
- Is topologically stable (genus = 1)

Parallel 5: Singularity Avoidance

DNA

Minimum radii prevent collapse:

- DNA helix cannot go below ~1 nm radius
 - Nucleosomes have fixed core diameter
 - 30-nm fiber has minimum bending
 - Too strong compression → DNA damage
- Reason:** Steric hindrance, Van der Waals radii, hydrogen bonds

T0

Minimum torus radius:

$$r_{\min} \approx 19.6 \ell_p \approx 3.2 \times 10^{-34} \text{ m} \quad (4)$$

Reason: Fractal dimension $D_f = 3 - \xi$ prevents singularity

Fifth Parallel**Both have fundamental lower limit**

	DNA	T0
Minimum radius	~1 nm	19.6 ℓ_p
Cause	Chemical	Geometrical
Consequence	DNA stability	No singularity

Parallel 6: Information Maximization**DNA****Problem:** 3 billion base pairs of information in ~6 μm **Solution:** Hierarchical folding**Result:**

- Information density: $\sim 10^9$ bits / μm^3
- One of the highest known information densities in biology
- Access when needed through local unfolding

T0**Problem:** Maximum physical information in finite space**Solution:** Fractal torus folding**Result:**

- Holographic principle: Information on surface
- Folding maximizes surface area
- Torus has maximum surface area for given volume

Sixth Parallel**Both maximize** $\frac{\text{Information}}{\text{Volume}}$ The folding is the **solution to an optimization problem.****Parallel 7: Compression Factor****DNA****Quantitative:**

Stretched DNA : ~ 2 m (5)

Chromosome : ~ 6 μm (6)Genome/chromosome : $\frac{2 \text{ m}}{6 \mu\text{m}} \approx 333,000$ (7)

This ratio compares the whole genome with a single chromosome. The compaction per chromosome (stretched ~4–5 cm) is **~10,000-fold**.

T0**Quantitative:**

$$\text{Planck scale : } 10^{-35} \text{ m} \quad (8)$$

$$\text{Hubble scale : } 10^{26} \text{ m} \quad (9)$$

$$\text{Orders of magnitude : } 61 \quad (10)$$

With $\xi = 1.33 \times 10^{-4}$: Scaling factor $\sim 1/\xi \approx 7500$ per level.

Seventh Parallel

Both achieve enormous compression without information loss

DNA: 10,000-fold (6 levels)

T0: $10^{61} \approx 7500^{16}$ (about 16 levels)

Parallel 8: Fractal Self-Similarity**DNA****Self-similar structure:**

- Helix (Level 1) → winds into solenoid (helix of helices, Level 3)
- Nucleosomes (tori, Level 2) → arranged on helix (Level 3)
- 30-nm fiber → folds into loops (Level 4) → into chromatin (Level 5)

Each level is a folded version of the previous one.

T0**Strict self-similarity:**

$$\frac{R_{\text{Level } n+1}}{R_{\text{Level } n}} = \frac{1}{\xi} \approx 7500 \quad (11)$$

In this model, the ratio R/r remains constant across scales.

Eighth Parallel

Fractal repetition of the same pattern

DNA: Qualitatively self-similar (helix → solenoid → loops)

T0: Quantitatively self-similar ($D_f = 3 - \xi$, fixed scaling ratio)

Parallel 9: Topological Stability**DNA****Topological invariants:**

- **Linking number** (Lk): Number of intertwinings
- **Twist** (Tw): Local turns
- **Writhe** (Wr): Supercoiling

Fundamental relationship:

$$\text{Lk} = \text{Tw} + \text{Wr} \quad (12)$$

These numbers are **topologically invariant** – they change only through cutting.

T0

Topological quantum numbers:

- Winding number $w = n_\phi/n_\theta$
 - Flux quantization $\Phi = n \cdot h/e$
 - Charge, spin, color charge from topology
- These are **topologically protected** – they change only at phase transitions.

Ninth Parallel

Topological stability

Both use **topological invariants** for stability:

DNA: Linking number preserves structure

T0: Winding number preserves quantum numbers

Parallel 10: Dynamic Unfolding

DNA

Unfolding when needed:

- **Transcription:** Local unfolding for RNA polymerase
 - **Replication:** Complete unfolding during S-phase
 - **Recombination:** Temporary unfolding for repair
 - **Regulation:** Acetylation → loose structure → accessibility
- The compaction is **reversible** and **regulatable**.

T0

Dynamic processes:

- Energy flows in torus variable
 - Torsion waves propagate
 - Particle creation = excitation
 - Phase transitions possible
- The structure is **static**, the energy **dynamic**.

Tenth Parallel

Static structure, dynamic processes

	DNA	T0
Structure	Chromosome (static)	Torsion crystal (static)
Dynamics	Local unfolding	Energy flows
Re-versible?	Yes	Yes (excitations)

.5 Why These Parallels?

Universal Optimization Problem

The Fundamental Question

Both biology (DNA) and physics (T0) face a **comparable challenge**:

How does one store maximum information (sequence / physical states) in minimum space without:

- Knotting (topology problems)
- Singularities (infinite energies)
- Information loss (entropy)
- Inaccessibility (must remain readable)

In both cases the answer is: **hierarchical torus folding with double helices**.

Mathematical Reasons

The following general reasons support the parallels:

1. Topology:

- Torus (genus = 1) is simplest non-trivial closed surface
- Enables two independent circulations
- Topologically stable

2. Geometry:

- Helix is natural curve in 3D
- Double helix maximizes stability
- Winding around a torus is a favorable arrangement

3. Information theory:

- Holographic principle: Information on surface
- Folding maximizes surface area
- Hierarchy allows logarithmic compression

Evolution vs. Fundamentality

An Interpretation

Did evolution "discover" torus geometry?

A possible answer [S]:

Evolution may have found this geometry because it is a **very favorable solution** to the information storage problem.

In the T0 description, physics also uses this geometry for fundamental structure.

DNA compaction would then be **not a random biological invention**, but an **expression of a general geometric principle**. Whether torus folding is really the only optimal solution is not proven in this document.

.6 Quantitative Comparisons

Compression Factors

System	From	To	Factor
DNA	~4–5 cm (stretched)	6 μm (chromosome)	~10,000×
T0	10^{-35} m (Sub-Planck)	10^{26} m (cosmic)	10^{61}

Table 1: Compression factors

Hierarchy Levels

System	Levels	Factor/Level	Geometry
DNA	6	~2–6×	Helix + Torus
T0	~16	~7500×	Torus + Fractal

Table 2: Hierarchical structure

Characteristic Lengths

DNA Level	Length	T0 Analog	Length
Double helix	2 nm	Sub-Planck	10^{-39} m
Nucleosome	11 nm	Particle	10^{-15} m
30-nm fiber	30 nm	Atom	10^{-10} m
Loop	300 nm	Molecule	10^{-9} m
Chromatin	700 nm	Macro	10^0 m
Chromosome	1 μm	Cosmic	10^{26} m

Table 3: Scale comparison (qualitative)

.7 Conclusion

Main Result

DNA compaction and T0 torus geometry show **ten structural parallels**:

1. Double helix / Double circulation
2. Winding numbers = quantum numbers
3. Hierarchical compaction
4. Toroidal geometry at each level
5. Singularity avoidance through minimum radius
6. Information maximization
7. Enormous compression factors
8. Fractal self-similarity

9. Topological stability
10. Dynamic unfolding

The parallels point to a **common geometric solution** for information storage; whether a common principle lies behind it remains an open question.

Summary Interpretation

Interpretation [S]

**In this reading, biology and physics use the same geometry
because it is a particularly favorable solution.**

DNA compaction can be read as a **biological expression** of the same **geometric principle** that, in the T0 description, also:

- Structures brain gyri
- Forms elementary particles
- Organizes the universe

In this interpretation, nature uses **the same solution on many scales** and **in different domains**:

Double helices → Tori → Hierarchical folding

(13)

This is a **common answer** to the problem:

Maximize information, minimize space, avoid singularities.